



Extended High-frequency Cues to Phoneme Recognition: Insights from ASR

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Abstract

There is emerging evidence that extended high frequencies (EHFs; >8 kHz) improve speech perception in noise. Yet, the mechanisms underlying this benefit remain unclear. We investigated whether EHF contribute to phoneme recognition using an automatic speech recognition (ASR) model. A neural network model was trained to decode phonemes from cochleagrams of broadband speech and speech low-pass filtered at 8 or 6 kHz in quiet and masked conditions with varying target-to-masker ratios (TMRs) and target-masker spatial separations. Compared with filtered speech, broadband speech improved phoneme recognition accuracy in masked conditions, particularly at lower TMRs, but showed no benefit in quiet. Removing EHF increased the probability of the model omitting a phoneme more for consonants than vowels. The findings suggest that the EHF benefit in adverse conditions may partly arise from enhanced phoneme processing, highlighting the potential of improving audiometry and ASR by including EHF.

Index Terms: extended high frequency, phoneme recognition, automatic speech recognition, cochleagram

1. Introduction

Although the human ear can hear up to about 20 kHz, there is a prevailing view that frequencies above 8 kHz, known as extended high frequencies (EHFs), are trivial for speech perception. Thus, routine audiological tests do not assess hearing at EHF [1] and phonetic analyses focus on bandwidths up to 4–8 kHz [2]. This view also underlies the development of automatic speech recognition (ASR) systems, which often rely on datasets with a 16-kHz sampling rate (e.g., LibriSpeech) or resample audio to 16 kHz, hence omitting EHF information.

However, there is growing evidence that EHF improve speech perception in noise (for a review, see [3]). Motlagh Zadeh et al. [4], for example, conducted a digits-in-noise task and measured listeners' speech reception thresholds, the signal-to-noise ratios at which they achieved 50% digit recognition accuracy. Results showed that providing access to EHF cues by low-pass filtering the noise at 8 kHz reduced reception thresholds by around 3 dB. EHF audiograms were also found to predict the listeners' subjective hearing difficulty in noise more accurately than audiograms at conventional frequencies (25–8 kHz). Similar EHF benefit was reported for more linguistically complex stimuli such as sentences [5].

A question that remains to be better understood concerns the mechanisms underlying the EHF perceptual benefit. One hypothesis is that EHF may directly convey cues relevant to phonemic processing. Support for this possibility comes from studies showing that phoneme recognition is possible using solely high-frequency cues or improves when such cues are

available [6, 7]. For instance, Vitela et al. [6] synthesized nonsense consonant-vowel tokens with speech content below 5.7 Hz replaced by noise. Even with such degraded stimuli, their listeners could identify phonemes well above chance, especially for consonants. On the other hand, reduced EHF hearing sensitivity may indicate poorer temporal processing in the cochlea [8, 9]. An alternative possibility is that EHF are a proxy for broader cochlear health. Thus, removing EHF does not eliminate critical phonemic information, but rather mimics speech perception under compromised cochlear function.

Whether better phoneme recognition is part of the EHF benefit is still controversial due to limitations of existing evidence, particularly in its generalizability to real-world speech perception. Support for identifying phonemes using only high frequencies sometimes comes from studies (e.g., [10]) using stimuli with lower-frequency cues completely removed, which are not representative of naturalistic speech. The high-pass filters used to isolate high frequencies could also have cutoff frequencies at 5–6 kHz (e.g., [6]), leaving it unclear whether accurate phoneme recognition was due to EHF above 8 kHz. Furthermore, the effect of EHF is modulated by aspects of the real-world auditory scenes such as the presence of an interfering talker and the spatial configurations of the talker. For example, due to high frequencies being more directional, EHF reduced speech reception thresholds more when the target and masking talkers were more spatially separated [11].

To address the limitations and assess whether EHF cues can in theory aid phoneme recognition, we experimented with an acoustic ASR model trained to decode phonemes from speech. Using ASR bypasses challenges in testing purely phonemic recognition in natural speech with human listeners, such as top-down lexical bias [12]. Also, EHF hearing thresholds vary greatly even among younger adults with normal hearing [4], making it difficult to control the cochlear health of the subjects. To provide a perceptually relevant model of human audition, we converted speech into cochleagrams, which simulate the non-linear frequency responses in the cochlea [13]. Cochleagrams have been shown to outperform traditional spectrograms in predicting brain responses to speech [14] and used to train convolutional neural network (CNN) models that capture the hierarchical auditory processing in the cortex [15]. These suggest the potential of cochleagram-based ASR for investigating speech perception.

We trained a CNN-BiLSTM model on cochleagrams of “broadband” (unfiltered) speech and speech low-pass filtered at 8 kHz (LP-8kHz) and 6 kHz (LP-6kHz) (Figure 1A). The 6 kHz cutoff was included as research on high-frequency speech cues also often tested frequencies above 5–6 kHz (e.g., [16]). In addition to speech in quiet, we simulated spatial listening in which target speech was masked by a competing talker at varying azimuth angles and target-to-masker ratios (TMRs). The broadband and low-pass filtering conditions were

compared to examine the impact of removing high frequencies on recognition accuracy. Additionally, we examined the impact on the model error rates of individual phonemes. As consonants have higher frequencies than vowels [2], it is expected that consonants would be affected more by removal of EHF.

2. Method

2.1. Dataset

Speech data were from the VCTK corpus [17], which includes 48-kHz recordings of sentences (e.g., *Please call stella*) produced by 109 English speakers. We used the 13,636 recordings from the 33 British-accented speakers and randomly split them into training (80%), validation (10%), and test (10%) sets. Word-level transcriptions were converted to phonemes using Phonemizer [18] with the espeak backend for British English. Phonemically non-contrastive vowel variations (e.g., [ʊɡən] ‘again’ vs. [bəlu:n] ‘ballon’) were merged, resulting in a 47-phoneme inventory (25 consonants, 22 mono/diphthongs).

2.2. Spatial audio synthesis

We synthesized spatial audio with the target speech masked by a competing talker from various directions (Figure 1B). Maskers were selected from non-British-accented speakers of the same corpus and paired with targets based on duration similarity (within -0.2 to $+0.8$ s). The shorter of the pair was zero-padded to match the duration of the longer and their intensities were adjusted to TMRs ranging from $+3$ to -12 dB SPL in 3 dB steps. Spatial separation between the two talkers was simulated using the CIPIC HRTF database [19], which provides head-related impulse responses (HRIRs) characterizing how sounds from different directions are filtered by the listener’s anatomical features (e.g., pinna size) before arriving at the eardrum. Each target-masker pair was assigned HRIRs from a randomly selected human subject of the database. Target speech was always convolved with HRIRs of both ears at azimuth $= 0^\circ$ (i.e., directly facing the listener) while the masker was convolved with those at θ . We tested $\theta = \pm 20^\circ$, $\pm 45^\circ$, $\pm 80^\circ$, and $\pm 120^\circ$. Target and masker signals were summed at each ear and the left and right channels were combined to form stereophonic audio. Target speech in quiet was processed using the same procedure but without a masker.

2.3. Filtering and cochleagram generation

Speech was either broadband (unfiltered) or low-pass filtered with a cutoff frequency of 6 kHz (LP-6kHz) or 8 kHz (LP-8kHz). Low-pass filtering was performed using a 1024-tap finite impulse response filter with a sharp roll-off [5]. Speech was transformed into cochleagrams using a gammatone filterbank mimicking the processing in the human cochlea [13] (Figure 1A). The filterbank comprised 128 filters with center frequencies spaced on an ERB scale from 20 Hz to 20 kHz and was applied separately to the left and right channels. The resulting cochleagrams underwent half-wave rectification and cubic root compression [20] and then were downsampled by averaging in 10-ms non-overlapping bins for computational efficiency. Cochleagram conversion was implemented using the brian2hears Python library [21].

2.4. Phoneme decoding model

A neural network was developed to decode phonemes spoken by the target talker from broadband or filtered cochleagrams.

Figure 2 illustrates the model architecture. The model consisted of a CNN block for spectro-temporal feature extraction and a bidirectional long short-term memory (BiLSTM) block for sequence learning. The CNN block included three 2D convolutional layers (3×3 kernels, 128/256/128 channels), each followed by batch normalization, ReLU activation, and dropout ($p = 0.2$). The output was flattened and passed to a three-layer BiLSTM (hidden size = 128, dropout = 0.2), followed by a fully connected layer with log softmax activation. The outputs were greedy decoded, selecting the most probable phoneme at each time step. No language model was applied. Accuracy was computed as $1 - \text{phoneme error rate (PER)}$, where PER is the Levenshtein distance, or the minimum number of substitutions, deletions, and insertions required to transform the actual phoneme string to the predicted one, normalized by sequence length. The model was trained using connectionist temporal classification loss and the AdamW optimizer (batch size = 32, max. 200 epochs, initial learning rate = 0.005, halved every 25 epochs). Early stopping (patience = 15 epochs) was applied to terminate training when validation PER no longer improved. The model was implemented in PyTorch (v2.1.2) and trained using NVIDIA Tesla A100 GPUs.

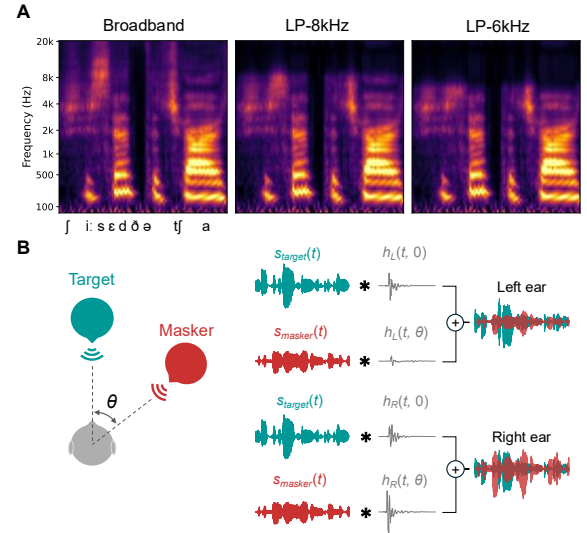


Figure 1: (A) Cochleagrams of broadband speech and speech low-pass filtered at 8 and 6 kHz. (B) Simulation of spatially separated target and masker using head-related transfer functions.

2.5. Analyses

All analyses were conducted on the test-set results. First, to assess how removing energy above 6 and 8 kHz affects phoneme accuracy in quiet, a linear mixed-effects regression model was fitted with the lme4 package [22] of R [23] including Bandwidth (reference level: broadband) as a fixed effect and target talker as a random intercept. For masked conditions, a mixed model was fitted including Bandwidth, TMR, the degree of target-masker separation (Separation), and all their interactions as fixed effects. Post-hoc comparisons assessed whether phoneme accuracy differed between broadband and either low-pass filtering condition at each level of TMR and Separation, with p -values corrected for false discovery rate.

Next, we further examined phoneme recognition in the presence of a masker. Several EHF studies use speech reception

thresholds to assess speech perception in noise [4, 11]. We followed a similar approach to calculate “phoneme recognition threshold.” Specifically, we fitted the same linear regression model for the masked conditions but including TMR as a continuous variable. The model coefficients were then used to construct an inverse function mapping an input accuracy rate to the TMR required to achieve that accuracy for a particular combination of bandwidth and degree of target-masker separation. Lower values for this TMR threshold indicate better performance. We tested input accuracy 60% to 80% and computed the threshold differences between LP-8kHz/LP-6kHz and broadband conditions. Positive values suggest threshold reduction due to access to high frequencies.

Finally, we explored how EHF affect the error rates of individual phonemes, focusing on consonants vs. vowels. We estimated the probability of a phoneme i (Ph_i) being replaced, deleted, or inserted using the following Bayesian formula:

$$P_k(E|Ph_i) = \frac{P_k(Ph_i|E)P(E)}{P(Ph_i)} \quad (1)$$

where E represents error type: either deletion, insertion, or substitution. $P_k(Ph_i|E)$ was estimated as the proportion of errors E that involved Ph_i in bandwidth condition k . $P_k(E)$ was calculated as the proportion of error of type E among all errors in condition k . $P_k(Ph_i)$ was the proportion of Ph_i of all phoneme instances. The resulting posterior probability controlled for the fact that phonemes differ in occurrence frequency in natural speech. We focused on the impact of EHF and computed the change in this probability from LP-8kHz to broadband for each phoneme, or $P_{LP-8kHz}(E|Ph_i) - P_{Broadband}(E|Ph_i)$. We then compared the change across consonant and vowel phonemes using independent t -tests. This error analysis was performed on the 28 phonemes with an occurrence frequency of at least 1% in the conditions with TMRs ≤ -9 dB, where the EHF effect was consistently observed (see Results).

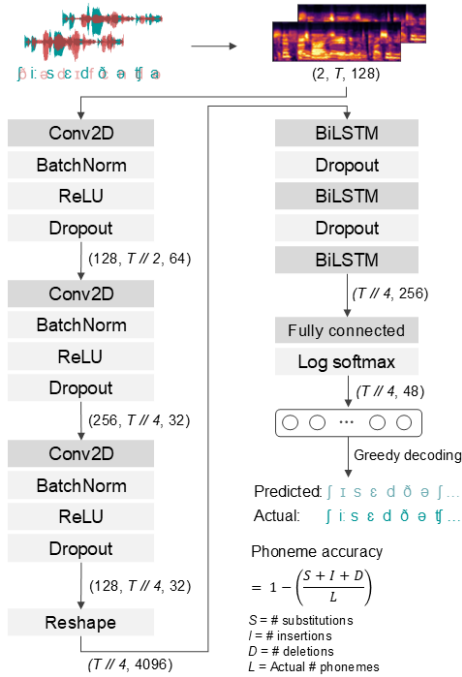


Figure 2: Architecture of the phoneme decoder and calculation of phoneme recognition accuracy.

3. Results

Figure 3 shows the mean phoneme recognition accuracy across all conditions. For speech in quiet, the broadband condition (mean = .872, SD = .097) did not differ significantly from LP-8kHz (mean = .876, SD = .094; $\beta = .004$, $t = 1.272$, $p = .203$) and LP-6kHz (mean = 0.874, SD = 0.094; $\beta = .002$, $t = .714$, $p = .475$). Hence, there was no evidence that frequencies above 6 and 8 kHz improve recognition in quiet above and beyond lower-frequency content. For masked speech, an ANOVA test of the mixed model indicated that all main effects and interactions were significant (all $ps < .001$). To more directly evaluate the bandwidth effects, the post-hoc comparisons were conducted and the significance outcomes are summarized with asterisks in Figure 3. Overall, accuracy was significantly higher in the broadband condition than LP-6kHz. A similar trend was found for broadband vs. LP-8kHz but it was more consistently seen at TMRs ≤ -9 dB. The findings suggested that high frequencies aid phoneme recognition when there is a masker, particularly at lower TMRs.

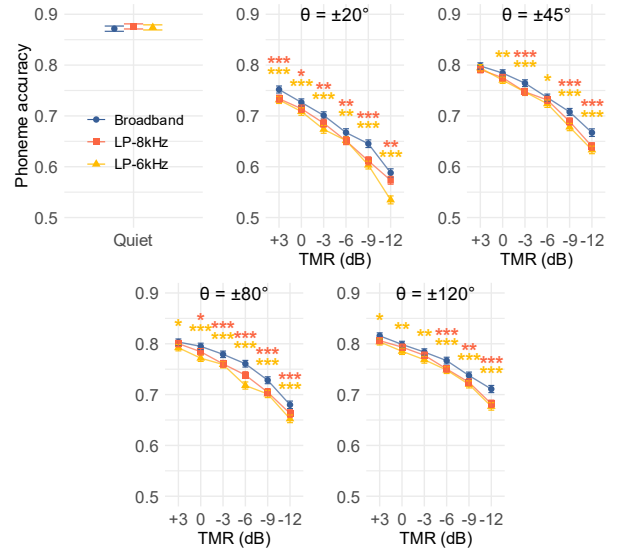


Figure 3: Mean phoneme recognition accuracy of quiet and masked speech by bandwidth condition.

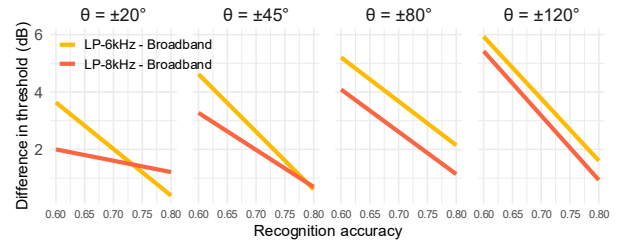


Figure 4: Difference in phoneme recognition threshold between LP-8kHz/LP-6kHz as a function of accuracy.

Figure 4 shows the difference in TMR threshold between LP-8kHz/LP-6kHz and broadband as a function of accuracy. The difference values were all positive and, in general, at least 1 dB, indicating that availability of high frequencies reduced phoneme recognition threshold. The reduction tended to be larger in the lower range of accuracy but became smaller for

higher accuracy, consistent with the minimal contribution of high frequencies in quiet from the accuracy analysis. Additionally, the tendency was more obvious as the degree of target-masker spatial separation became larger.

Figure 5 plots the change in the probability of each error type from LP-8kHz to broadband for each phoneme, computed using the results from all masked conditions with TMRs ≤ -9 dB. For deletion and substitution, the changes were mostly positive, indicating that excluding EHF increased error rates. Importantly, for deletion, the increase was significantly greater for consonants than vowels (Welch’s $t(25.919) = 3.303$, $p = .003$). The changes in probability did not differ significantly for substitution (Welch’s $t(24.743) = 1.655$, $p = .111$) and insertion (Welch’s $t(25.969) = -1.071$, $p = .294$). Note that /f, v/ showed the greatest increase in both deletion and substitution probability, which will be discussed in detail. We repeated the error analysis for the broadband vs. LP-6kHz contrast and found the same significance patterns, except this time the increase in substitution rate was also significantly higher for consonants than vowels.

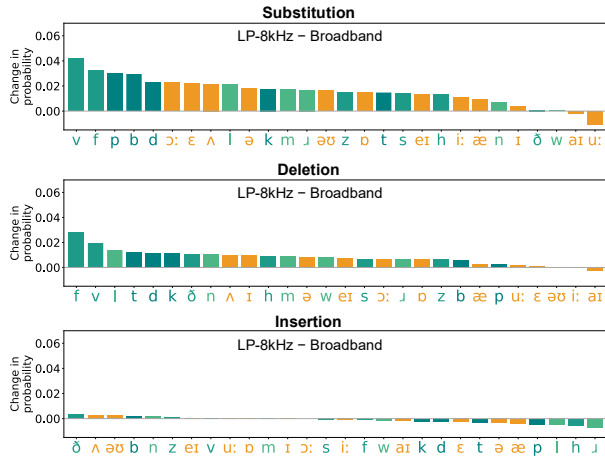


Figure 5: Change in the probability of substitution, deletion, and insertion error from broadband to LP-8kHz condition for each phoneme.

4. Discussion

This study investigated the contribution of EHF to phoneme recognition by training a cochleagram-based ASR to decode phonemes in quiet and in the presence of a masker with varying degrees of simulated spatial separation. In the masked conditions, broadband speech showed higher recognition accuracy compared with speech low-pass filtered at 8 or 6 kHz, especially at TMRs lower than -9 dB. Such difference was not found for the quiet conditions. Broadband speech also required lower TMRs to reach the same accuracy as filtered speech, with the threshold reduction increasing as accuracy decreased and target-masker separation widened. Error analyses revealed that removing EHF disproportionately increased deletion rates for consonants compared with vowels.

The results support the hypothesis that EHF contains information directly relevant to phoneme recognition. Therefore, the perceptual benefit of EHF in studies with human listeners [5, 24] can arise at least partly from improved decoding of phonemic content. This gain in bottom-up processing in turn lowers the cognitive resources needed to

recover ambiguous speech sounds using other cues, driving the correlation between EHF hearing sensitivity and subjective report of hearing in noise difficulty [4]. However, note that neither frequencies above 8 kHz nor those above 6 kHz further improved phoneme recognition in quiet. This aligns with the speech intelligibility index model [25], which evaluates the importance of different frequency bands for intelligibility in quiet and predicts minimal additional benefit for 7–10 kHz. These findings highlight the need to consider both optimal and suboptimal listening situations when evaluating EHF effects.

Consistently, the recognition threshold analysis revealed that the threshold reduction due to high-frequency cues was as small as around 1 dB in the higher-accuracy range but increased to 2–6 dB when accuracy was lower. The trend was more obvious when the target and masker were further apart, similar to the patterns observed in previous work [11]. This is presumably because the interference from the masker would be smaller at the ear contralateral to the masker source than the ipsilateral ear particularly for higher frequencies [26, 27]. Such an interaural difference enlarged as the target-masker separation became larger, providing the model more opportunities to glean additional phonemic information at high frequencies using the input from the contralateral ear.

The increase in deletion probability from broadband to LP-8kHz suggests that, as expected, consonants are more susceptible to removal of EHF than vowels. Furthermore, we observed that the deletion and substitution increased to the largest for the labiodental fricatives /f, v/. This pattern was strikingly similar to the behavioral results from Polspoel et al. [5], who also found that recognition of /f, v/ in noise by human listeners were impaired the most among all phonemes when EHF energy was filtered out. Compared with other consonants, /f, v/ are relatively soft sounds with a flat spectrum and a mean spectral peak close to 8 kHz [28, 29]. The low-pass filter at 8 kHz can eliminate a considerable proportion of the energy in these fricatives, causing the ASR model to omit them.

The EHF benefit for human listeners has led to recommendations for including hearing thresholds above 8 kHz in standard audiometry [1]. An implication of our experiment with the cochleagram-based ASR phoneme decoder is that EHF hearing sensitivity may serve as a predictive marker of the listener’s bottom-up phonemic processing ability in adverse listening situations. The current findings also suggest reconsideration of EHF in designing ASR systems, especially for applications to real-world auditory environments with voices coming from various directions. It would be of interest to examine if the EHF benefit extends to larger-scale experiments testing state-of-the-art ASR models (e.g., wav2vec 2.0 [30]) on larger datasets with high-quality audio sampled at >40 kHz (e.g., Common Voice [31]).

One limitation of this study is that the simulated spatial listening does not always resemble realistic cocktail party problems, in which interfering talkers may be facing away rather than directly in front of the listener. As EHF effects are shown to be more pronounced with non-facing maskers than facing maskers [32], the contribution of EHF to phoneme recognition might be underestimated with the spatial setup used here. It would thus be useful to experiment with speech from talkers with various head orientations [33]. Additionally, we did not test the impact of EHF on other aspects of speech perception (e.g., sound quality judgment). Investigating these aspects is crucial for a more comprehensive understanding of the mechanisms underlying EHF perceptual benefit.

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6. References

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